

BLACK BOX INSTALLATION EXAMPLE LIEBHERR LTM CONSTRUCTION YEAR > 2018



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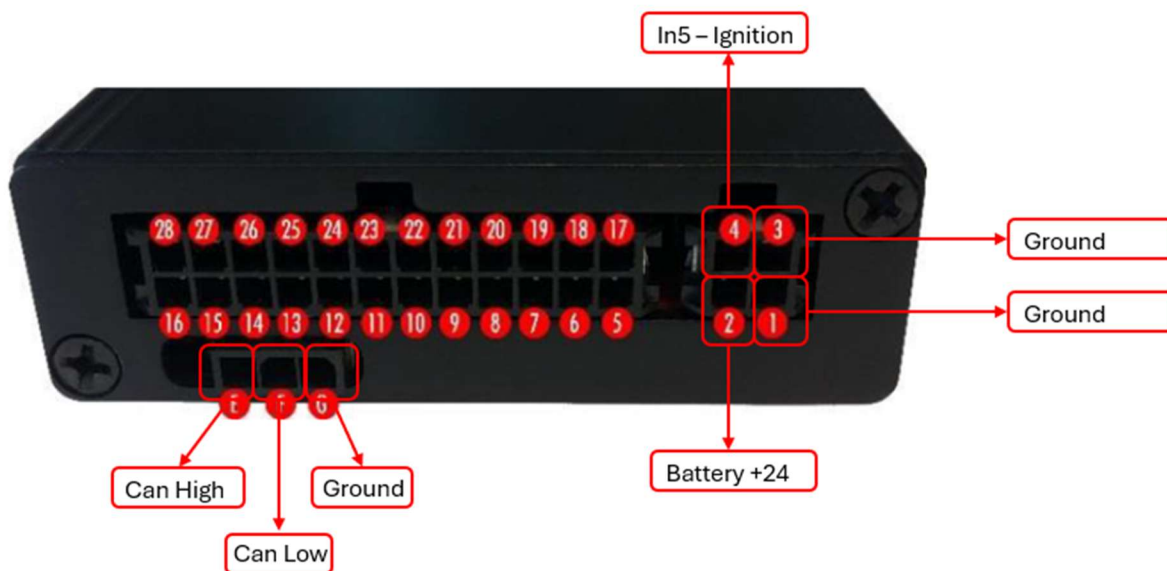
Group FLEET
Director

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1. Connection overview

Only shown the connection for this installation example.

Pin	Signal	Description	Crane connections
1	GND for VCC	Ground for VCC – Chassis ground	Connected to chassis ground – MSP1
2	VCC	24V – Battery supply	Connected to +24 v BEFORE master switch – X50
3	GND for I/O	Ground for I/O	Joined with pin 1 and Can Grounds – MSP1
4	In5	Digital/Analog Input 5 – Ignition signal	Ignition – Key contact – F37 to F48 possible
E	Can 1 - CAN_H	CAN H	X4 pin 14 BTB or X72.pin 5
F	Can 1 - CAN_L	CAN L	X4 pin 13 BTB or X72.pin 6
G	Can 1 - GND for CAN	Ground for CAN – Same as Chassis ground	Joined with pin 1 and Can Grounds – MSP1



LoRa	External LoRa antenna
RF	External 2.45GHz. antenna
GPS	External GPS antenna
GSM	External GSM/GPRS antenna

2. Package content – Extended

Following materials will be delivered from HQ:

- Extended Black box – sap 10094697.
 - o Set of wiring :
 - 1x CANBUS harness, I/O harness and supply harness.



- Antenna – For this installation the antenna will be an inside type.
 - o Internal – sap 10094701



3. Tools and requirements

Standard technician – electrician tools are needed to perform the installation, the manual wont specified which set of standard tools the installer should use, but give an overview of the “non common” tools which are required.

- Set of electrical pin extractor tools. Although in most cases this will be a standard kit for any BU, it might defer from a standard tool. For hooking/joining the signal cables from the black boxes to the crane wiring.



- P-can viewer usb dongle – sap 10067404. Dongle used for the installation software tool.



- Crimping pliers.

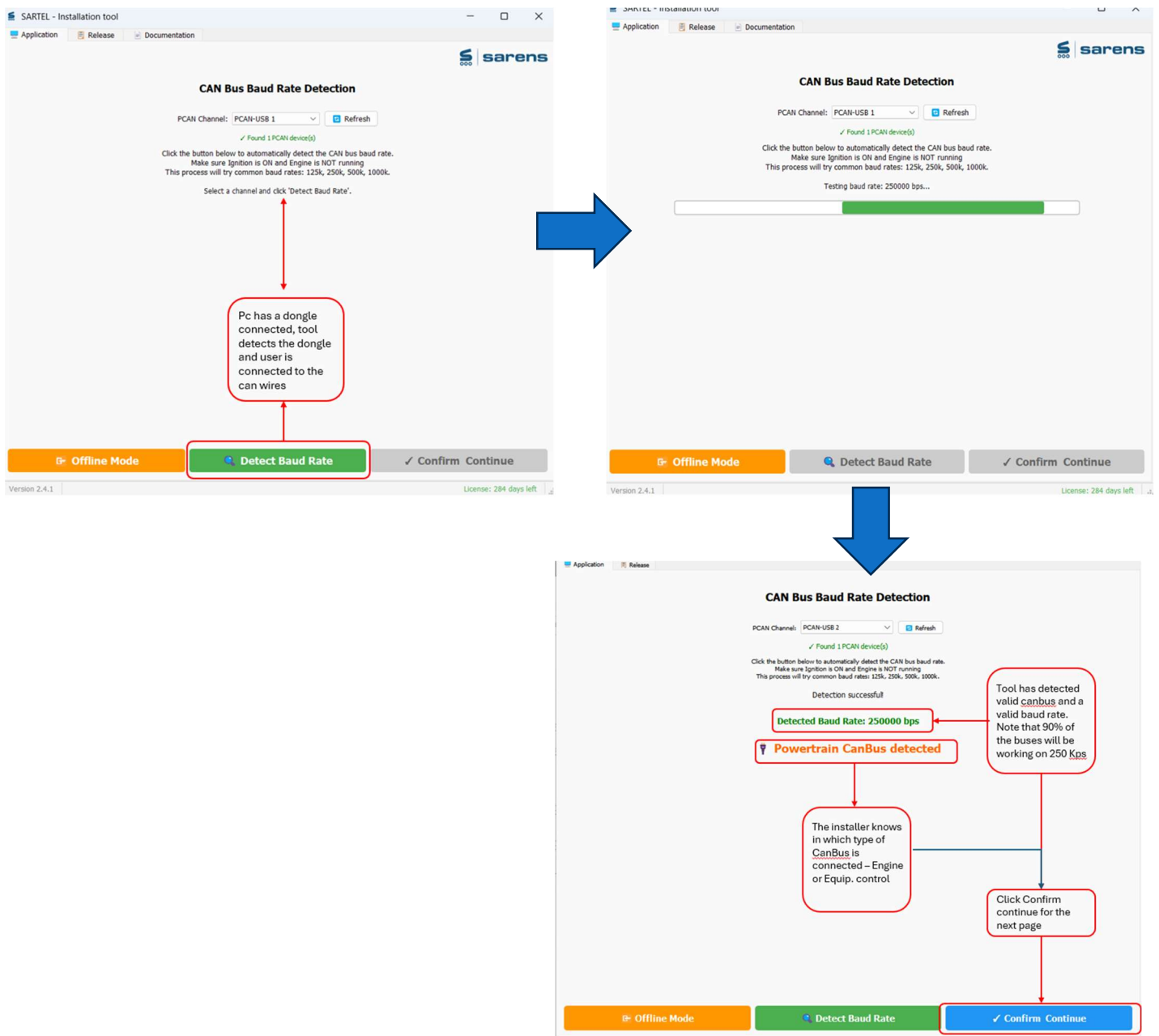


4. Pre-installation check.

* **Liebherr crane as from 2018** *. As from 2018 production Liebherr cranes are equipped with a FMS connector (Fleet management system). This simplifies the connection, allowing all signals needed for our Black box being fetch from the same plug.

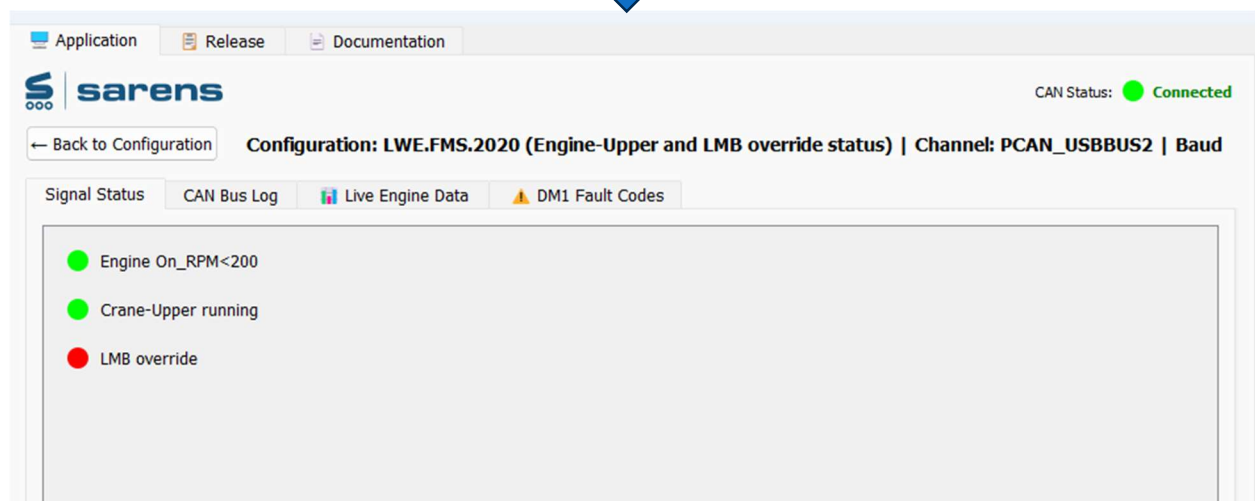
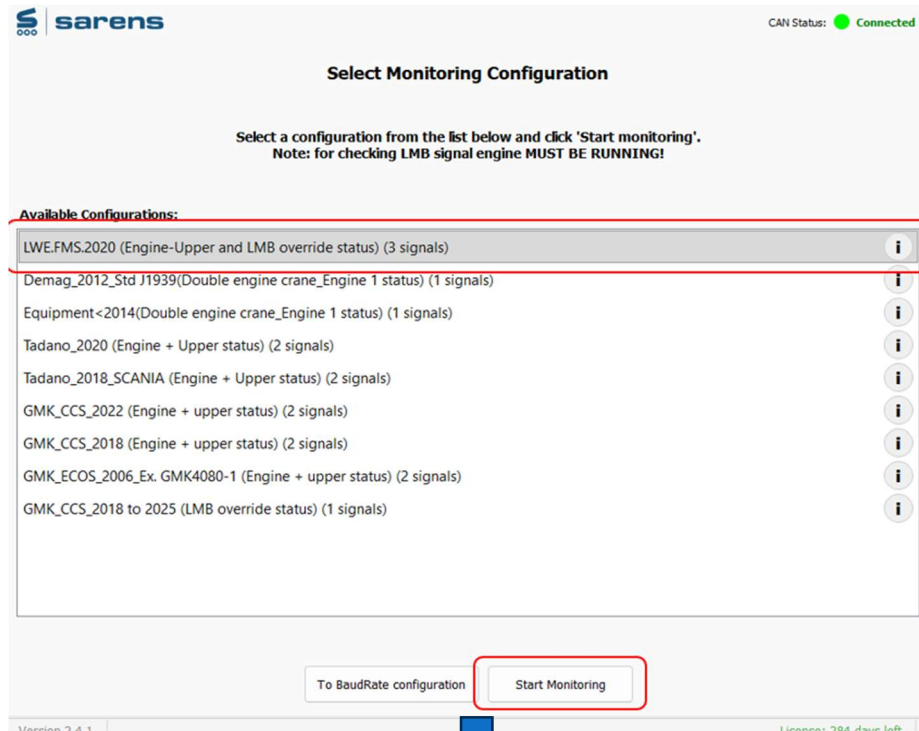
- Single BB
- BB-Can1 connected to the FMS connector.

Connect the Pcan dongle with the loose pins to the BTB X4.14 (CanH) and X4.13 (CanL), open up the Sartet tool and search for the baudrate.



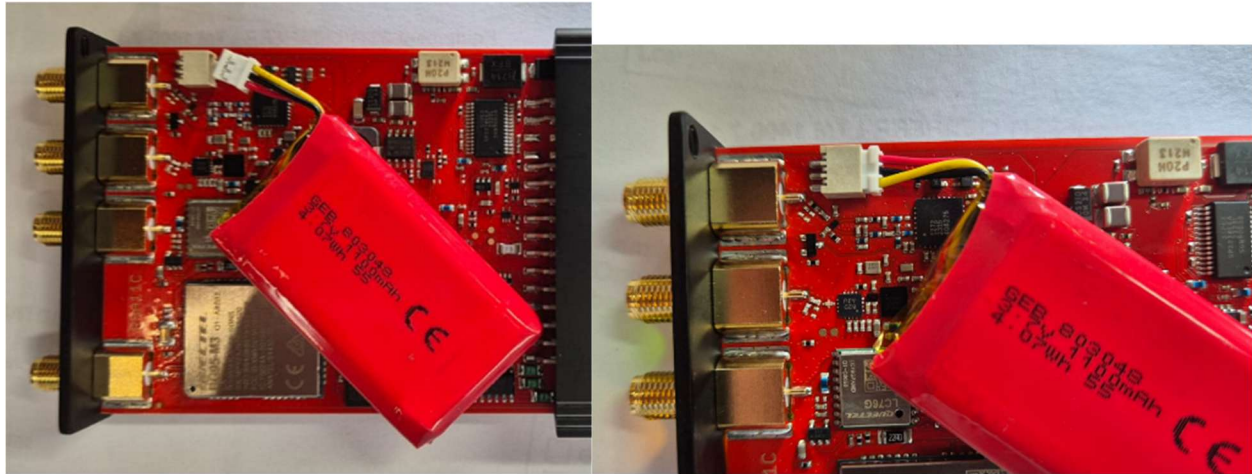
Select the LWE.FMS.2020 configuration and check that the require signals for the Black box are present:

- Engine running carrier.
- Crane running / upper structure on.
- LMB override.



5. Hardware Installation.

Open the BB and connect the battery:

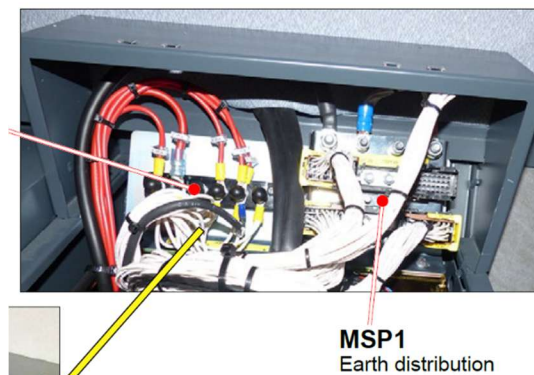


Liebherr crane as from 2018. Since 2018 Liebherr integrates in production a FMS connector (Fleet management system). This connector allows us fetching all carrier, engine and crane operation data.

The connection points for this Canbus is located on the BTB's. See BMK information or example installations for more specific information. Generally the BTB is installed on the carrier middle console for LTM cranes, and electrical cabinets for lattice boom cranes (LBC).

Connect the following wires from the BB to the below points in the crane :

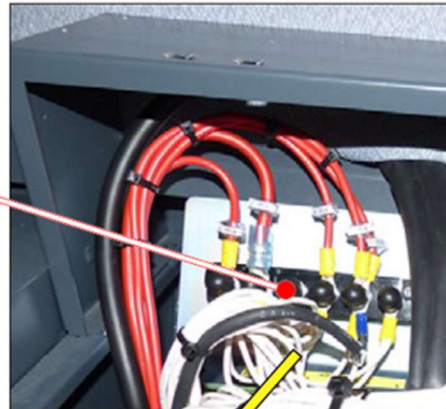
- **Pin 1 – Pin 3 – Pin G** → Ground.
 - o MSP 1



- **Pin 2** → Battery supply.
 - o Connect it to X50 next to the ground supply.

X50

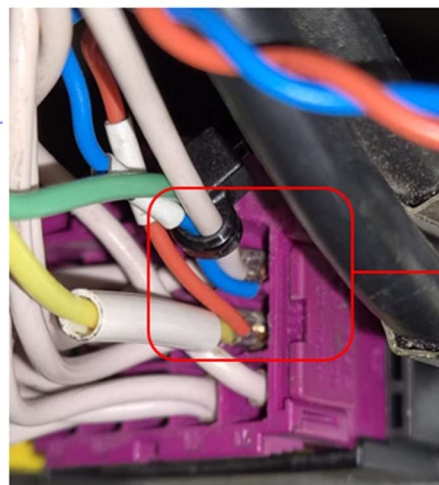
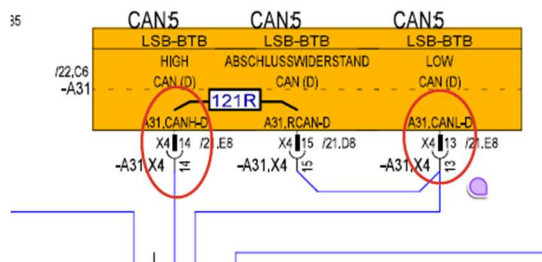
Clamping strip (distributor battery voltage)



- **Pin 4** → I/O type In5. Ignition Signal.
 - o Connect it to a fuse. Fuses between F37 to F48 should have available ignition signal. * to be checked in between models.
- **CanBus plug 1:**
 - o Pin E → Can H to X4.14
 - o Pin F → Can L to X4.13

Can Connection – when FMS is available there will be a yellow and green cables on the plug X4 – pin 14(Can H) and pin 13 (Can L).

Connect Can wires from the BB (Can L to 13 and Can H to 14) as shown on the pictures below.



X4 BTB
Pin 13 CanL
Pin 14 CanH

- Antenna installation.
 - o Route the two antenna wires under the dashboard and place the antenna as close as possible to the dashboard. Degrease the window before sticking it.



6. Installation sheet.

At the end of the installation the technician/installer must fill up the installation sheet (templates available in Sharepoint), take the required pictures – as specified in the installation sheet and forward a copy to xxxx@sarens.com.

The installation sheet will be uploaded to SarPlan under the crane resource number.

For this installation the **D- BB connection sheet_SUIVO extended_Singel engine** should be used!

The fleet manager/fleet team is responsible for providing the HQ via feel.tracking@sarens.com the below documentation:

- Installation sheet properly filled.
- Installation pictures (antenna location + BB installation location).
- BB Imei picture.

Following boxes should be filled for this installation example:

- General crane identification:

Crane number	Brand /Type	Date	Supplier
		Technician	SUIVO / GFS
Serial number crane	IMEI number Black Box		Maintenance order (SAP)

- Crane specific. For LTM cranes above 2020:

Connection for supplies:								
	Pin	Naming	Data type	Info page schemas	Wire color	Crane connector name (From schemas)	Signal checks	Signal function on Platform
	2	24V	KL 30 - Battery supply				Supply from battery KL 30+. Before master switch! Write voltage value:	Power supply for the tracker when the machine is not in used
	1	Ground	KL 31 - Chassis ground				Ground / masa connection	Ground
	3	Ground	KL 31 - Chassis ground				Ground / masa connection	Ground
	4	Ignition signal	KL 15 - Ignition signal				Ignition signal KL+15. Note that BB needs to have ignition signal for data collection. Write voltage value:	Wake up signal for CanBus data collection - ⚠ The Canbus data recording will start when the signal is detected. Ex. Single engine- The BB is installed on the US, the ignition signal used for BB should be high when the carrier or crane ignition are on (see crane connection examples)

Detection successful
Detected Baud Rate: 250000 bps
Example: 250 Kbit/s

 **Powertrain CanBus detected**

Other crane brand or Liebherr older than 2018:								
CAN 1 or I/O		Engine data	CanBus - Connected to the engine can		Can H :	Installation tool - Sartel- check & note Baudrate:	Engine running status. Hour counter function. Engine data values collection if CanBus connected. Diagnostic engine if CanBus connected.	
	E	CAN H	☒		Can L :	Installation tool - Sartel- check & note		
	F	CAN L				Configuration:		
	24	Digital In 1	I/O standard D+ ☐			Installation check with multimeter Write Voltage va		

Available Configurations:
LWE.FMS.2020 (Engine-Upper and LMB override status) (3 signals)

Example:
LWE.FMS.2020

- Crane hours:

Information working hours	
Working hours Primary Engine / Lower Engine :	
Working hour Upper:	
Travelling (crawler):	
Specified position of BB in the crane:	
Travelling (crawler):	
Take pictures of the following:	
IMEI sticker:	☒
Location of BB in the crane:	☒
Connection of Can 1:	☒
Connection of Can 2 (if connected):	☐
Engine id plate:	☒

Once completed the installer should forward the pictures and connection sheet. The fleet manager is responsible to send the information to fleet.tracking@sarens.com. Once received the Black box team will configure the box over the air, it will then become visible on the platform and data will be uploaded on AX.

